

Computer Science & IT

Database Management System



Transactions & concurrency control

Lecture No. 08

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Recap of Previous Lecture

Topic

[View Serializable Schedule](#)



Topics to be Covered



✓
Topic

Concurrency control protocols

✓
Topic

Simple use of shared and exclusive locks

✓
Topic

Two phase lock (2PL)



Topic : Concurrency Control Protocols

- ✓ • Concurrency Control Protocols are responsible for avoiding the execution of non-serializable schedules.
- ✓ • Concurrency Control Protocols are suggested to ensure serializability



Topic : Concurrency Control Protocols

- There are two types of concurrency control protocols.

✓ 1. Lock based Protocols ✓

★ Two Phase Locking Protocol (2PL)

- ✓ (i) Basic 2PL
- ✓ (ii) Strict 2PL
- ✓ (iii) Conservative 2PL
- ✓ (iv) Rigorous 2PL

✓ 2. Time-stamp based Protocols

- (i) Basic Time stamp ordering Protocol.
- (ii) Thomas write Time-stamp ordering Protocol



Topic : Lock Based Protocols

- Lock is a variable associated with a data item that describes status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. **Shared LOCK [S]**: It is read only lock.

If transaction wants to perform only read opⁿ on a dataitem, then transaction will request for a shared lock on that dataitem

2. **Exclusive LOCK [X]**: It is read as well as write lock.

T
~~R(A)~~
We are not allowed to perform read or write opⁿ directly

T
Requesting for shared lock on dataitem 'A'
~~S(A)~~ ← "Let granted"
R(A) ✓ {Allowed}
~~W(A)~~ not allowed, transaction T does not have write permission w.r.t. dataitem 'A'
~~R(B)~~ Not allowed, dataitem 'B' is not locked by transaction T.



Topic : Lock Based Protocols

- Lock is a variable associated with a data item that describes status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. **Shared LOCK [S]:** It is read only lock.

T
~~W(A)~~
Not allowed

2. **Exclusive LOCK [X]:** It is read as well as write lock.

{ Transaction can request for an Exclusive lock
On dataitem A Only if transaction wants to perform
at least one write opⁿ on dataitem 'A' }

Requesting an Exclusive lock on dataitem 'A'

T
X(A) ← let granted
R(A) ✓ (Allowed)
W(A) ✓ (Allowed)
S(B) ✓
R(B) ✓ (Allowed)
~~W(B)~~ Not allowed
~~R(C)~~ - / -
~~W(C)~~ - / -



Topic : Lock Compatibility Table

Locks held by Transaction T_i

data item 'A'	locks held by transaction T_i		
	Shared lock 'S'	Exclusive lock 'X'	
locks requested by transaction T_j	Shared lock 'S'	(✓) Allowed	Not allowed
	Exclusive lock 'X'	Not allowed	Not allowed

Note:-

To ensure serializability a non-serializable schedule must not be allowed to execute.



Topic : Simple use of shared and exclusive locks

ie, No restriction { ie, data items can be locked/unlocked }
in any order at any time

Consider the following Schedule.

	T ₁	T ₂
✓	X(A)	
✓	R(A)	
✓	W(A)	
Unlock(A) ≡	U(A)	S(A) ✓
		R(A) ✓
		S(B) ✓
		R(B) ✓
		U(A)
		U(B)
✓	X(B)	
✓	R(B)	
✓	W(B)	
	U(B)	

Given schedule
is a non-serializable
Schedule.

But it can be allowed
to execute using
Simple use of
Shared and Exclusive
lock.

A non-serializable
Schedule may be allowed
to execute,
∴ Simple use of Shared
and Exclusive locks
does not ensure
serializability.



Topic : Basic Two Phase Locking Protocol

→ There are two phases in Two phase locking protocol

- ✓ 1. Growing phase / Lock acquiring phase
- ✓ 2. Shrinking phase / Lock releasing phase

→ In Two Phase Locking Protocol, transaction T is allowed to request for a lock ^{on any data item} only if transaction T has not performed any unlock operation.

eg:

Transaction (T)

Lock acquiring
Phase

(or)

Growing Phase

S(A)
R(A)
X(B)
R(B)
X(C)
S(D)
W(B)
R(C)
R(A)

U(A)

R(D)

W(C)

U(C)

S(A)/X(A)

S(E)/X(E)

W(B)

U(B)

U(D)

Shrinking Phase
starts

Shrinking Phase
(or)
Lock releasing Phase

not allowed

not allowed

① Q. Check whether the schedule is allowed to execute using 2PL or not?

⑤

	T ₁	T ₂
1	X(A) R₁(A) W₁(A)	
		S(A) R₂(A)
		R₂(B)
	R₁(B) W₁(B)	

it will be denied because 'A' is already locked by T₁ in Exclusive mode

If a lock request is denied, then that transaction will go into a time-out period and it will request for the same lock once time-out period is over.

① Q. Check whether the schedule is allowed to execute using 2PL or not?

⑤

	T ₁	T ₂
✓	X(A) R ₁ (A) W ₁ (A) U(A)	S(A) ✓ R ₂ (A) S(B) R ₂ (B) U(A) U(B) -
	 X(B) R ₁ (B) W ₁ (B)	

denied
because T₁
has already
performed unlock
op

① Q. Check whether the schedule is allowed to execute using 2PL or not?

⑤

	T ₁	T ₂
✓	X(A) R ₁ (A) W ₁ (A) X(B) U(A)	
		S(A) ✓ R ₂ (A) S(B) denied, 'B' is already locked by T ₁ R ₂ (B)
	R ₁ (B) W ₁ (B)	

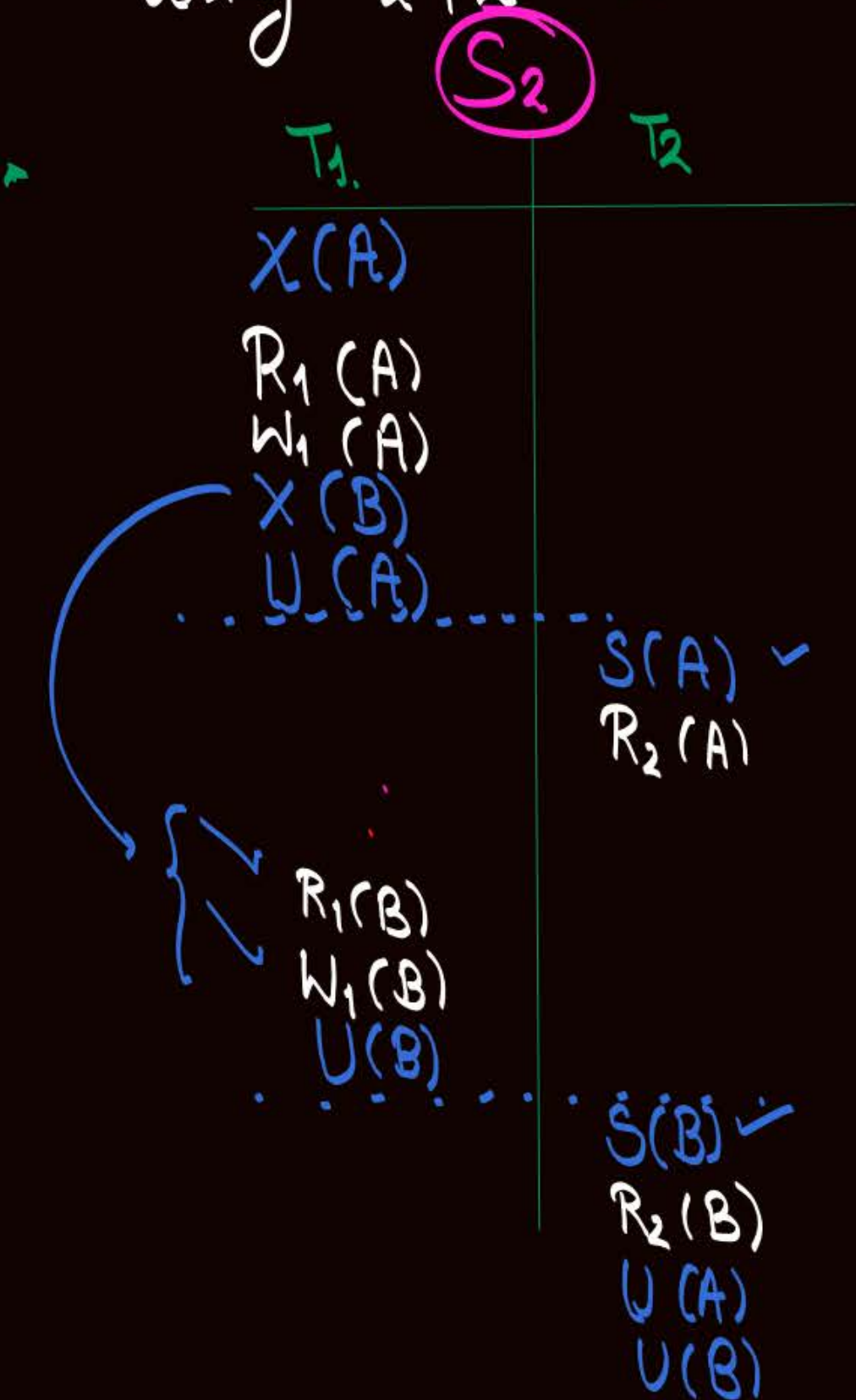
① Q. Check whether the schedule is allowed to execute using 2PL or not?

⑤

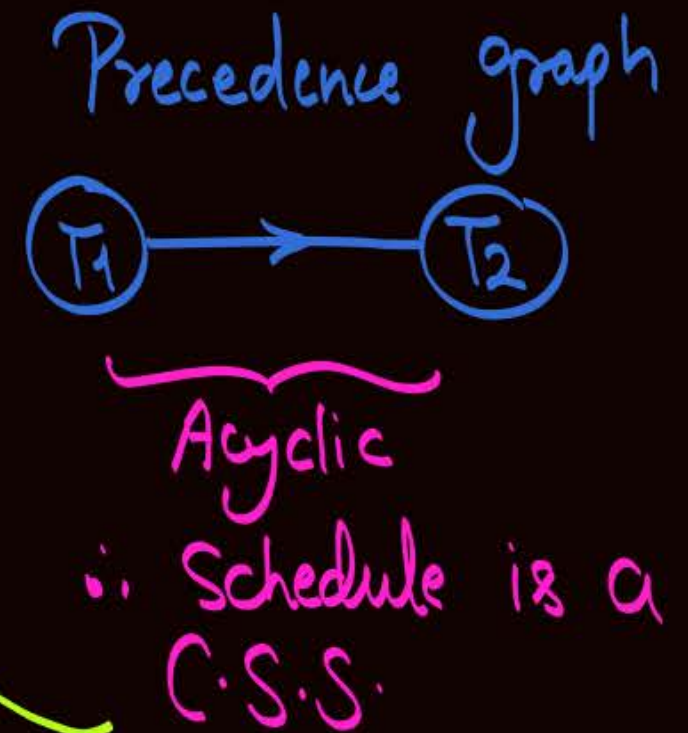
T_1	T_2
$R_1(A)$ $W_1(A)$	$R_2(A)$ $R_2(B)$
$R_1(B)$ $W_1(B)$	

→ Schedule is a non-serializable schedule and it is not allowed to execute using "2PL" protocols.

② Q. Check whether the schedule is allowed to execute using 2PL or not?



It is a Conflict Serializable Schedule and it is allowed to execute using "2PL protocols"



③ Q. Check whether the schedule is allowed to execute using 2PL or not?

S₃

Until transaction T₂ performs all the operation on data item 'B', it will not unlock 'B'.

T ₁	T ₂
	X(B)
	R(B)

S(A)
R(A)

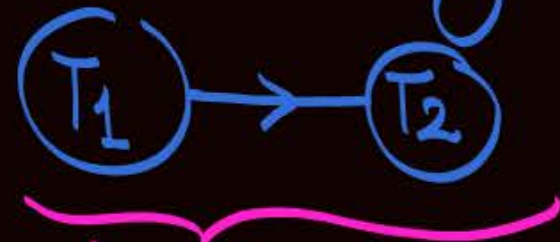
~~S(B)~~
R(B)

∴ this S(B) can never be granted at this position
∴ Not allowed by 2PL

W(B)

Schedule is a C.S.S., but it is not allowed by "2PL"

Precedence graph



Acyclic
∴ it is a C.S.S.

④ Q. Check whether the schedule is allowed to execute using 2PL or not?

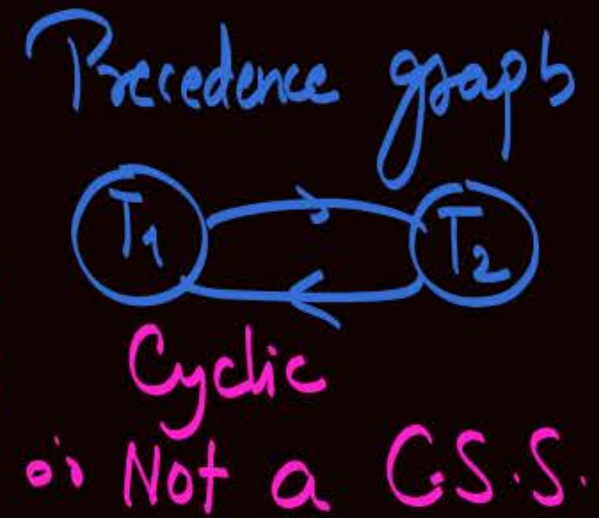
S₄

T ₁	T ₂
X(B) R(B)	
	X(B) W(B)
W(B)	
	W(B)

~~X(B)~~ denied
↳ can not be fulfilled
∴ Not allowed by 2PL

It is not a CSS;
but it is a View
Serializable schedule

⇓
It is not
allowed by 2PL



But
 $S_4 \equiv T_1 \rightarrow T_2$
∴ S₄ is a
Serializable
Schedule
(i.e. View
Serializable)



Topic : Basic Two Phase Locking Protocol

1. 2PL ensures serializability, i.e., a non-serializable schedule is never allowed to execute using 2PL.
2. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule, but every conflict serializable schedule need not be allowed to execute by 2PL.
 { We can also say 2PL ensures Conflict serializability }
 { i.e. a non-C.S.S. is never allowed by 2PL }
3. If schedule is not a conflict serializable schedule, then that schedule is not allowed to execute using 2PL.
4. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule and conflict equivalent serial schedule can be given by order of lock points.

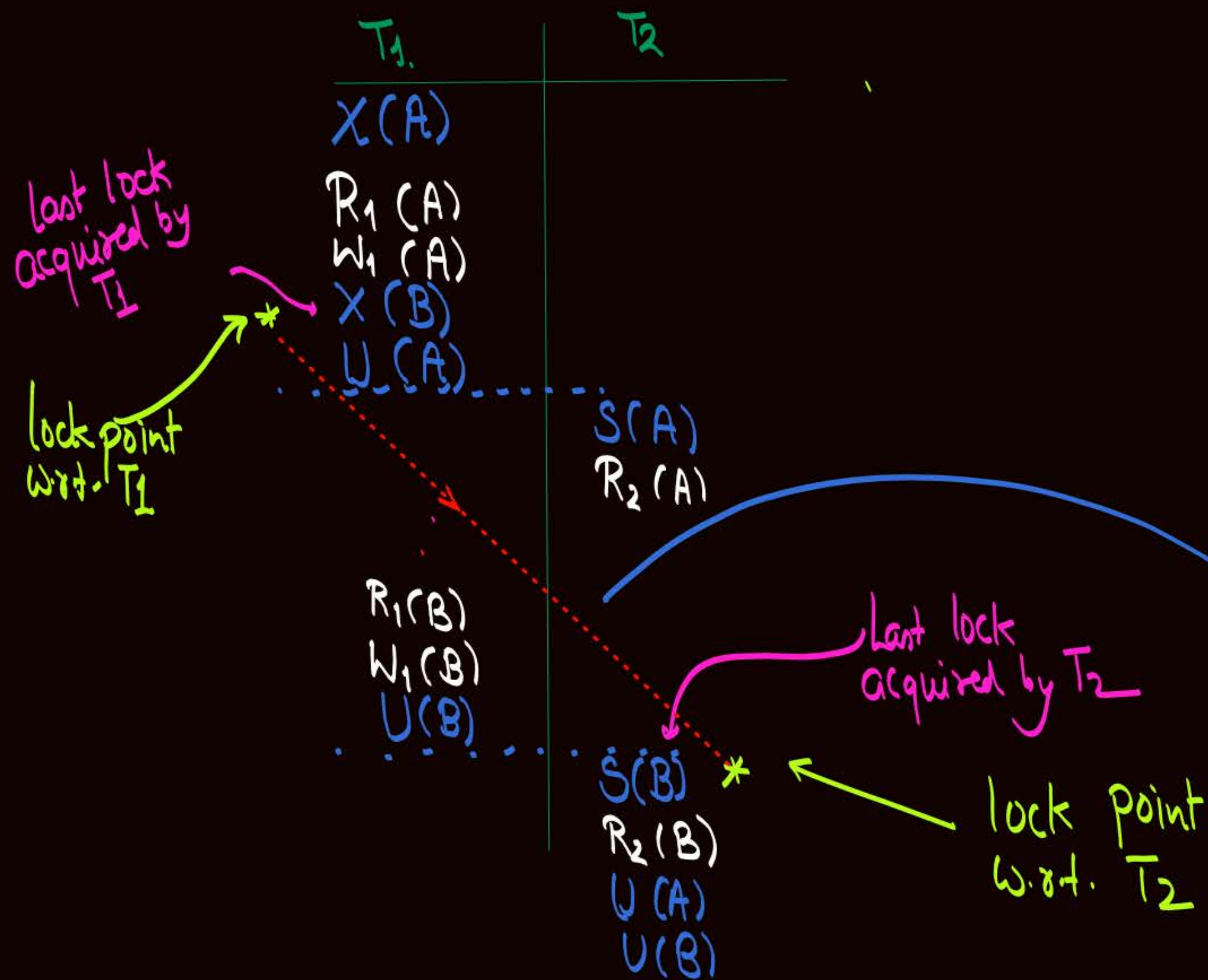


Topic : Lock Point



- The Point at which the growing phase ends, i.e., the point at which transaction takes the final lock it needs to carry on its work. is defined as lock point w.r.t. that transaction.

eg.



It is a Conflict serializable Schedule, and it is allowed to execute using "2PL protocols"

Conflict equivalent serial Schedule w.r.t. order of lock point is

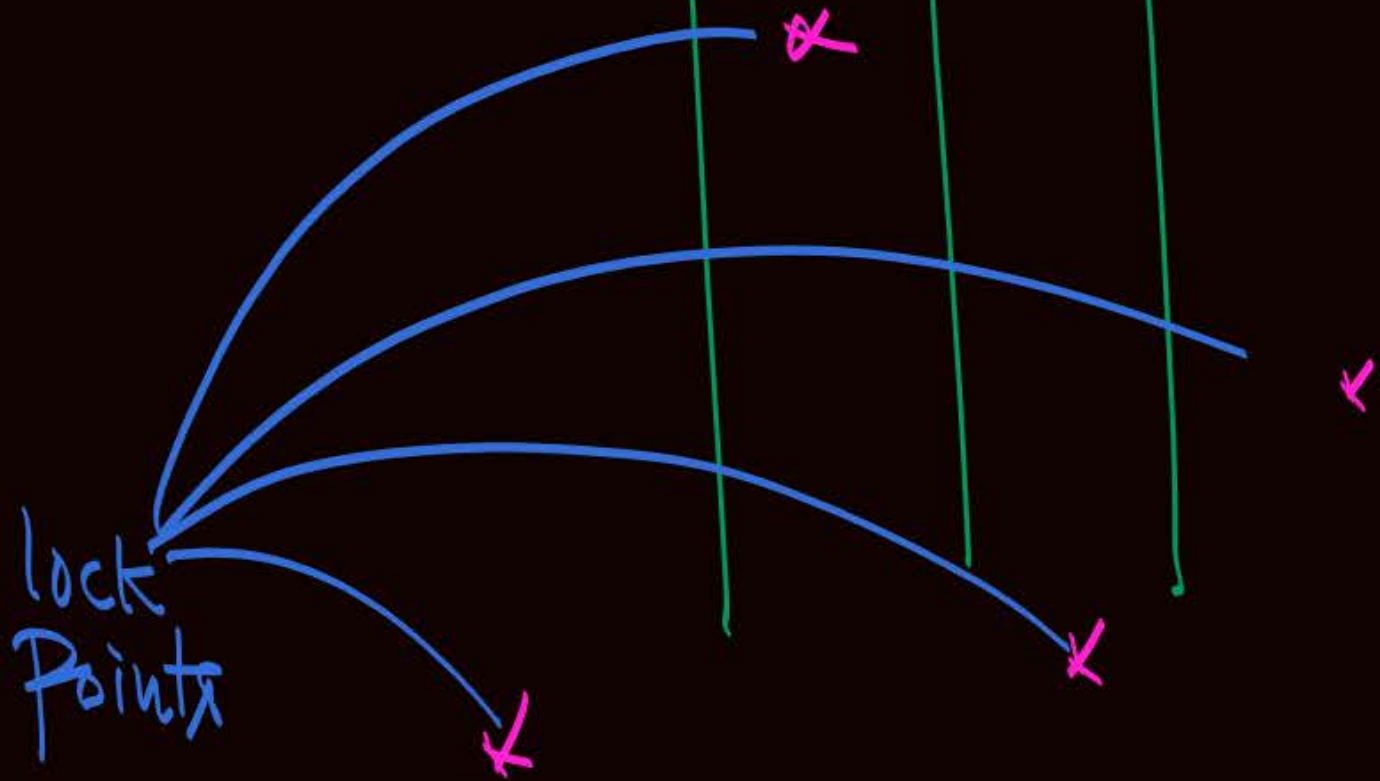
T₁ → T₂

eg. Lock Points,

Consider the following

'S'

T ₁	T ₂	T ₃	T ₄



Schedule 'S', and assume that
Schedule 'S' is allowed by 2PL,

then conflict equivalent serial
Schedule based on order of
lock point is

$T_2 \rightarrow T_4 \rightarrow T_3 \rightarrow T_1$



2 mins Summary



✓
Topic

Concurrency control protocols

✓
Topic

Simple use of shared and exclusive locks

✓
Topic

Two phase lock (2PL)

✓ **THANK - YOU**